

AUTONOMOUS FRAMEWORK FOR REAL-TIME DATA QUALITY, PRIVACY, AND SEMANTIC INTEGRITY IN STREAMING ENVIRONMENTS

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AUTONOMOUS FRAMEWORK FOR REAL-TIME DATA QUALITY, PRIVACY, AND SEMANTIC INTEGRITY IN STREAMING ENVIRONMENTS



REAL-TIME DATA QUALITY ASSESSMENT AND EXPLAINABLE PREPROCESSING FRAMEWORK WITH BENCHMARKING AND ROLLBACK SUPPORT FOR STRUCTURED DATA

Bachelor of Science (Hons) in Information Technology
Specializing in Data Science

Real-Time Data Quality Assessment and Explainable Preprocessing Framework with Benchmarking and Rollback Support

❖ Proven Gap:

- Current real-time data tools either **detect problems** (missing values, duplicates, anomalies) or **clean data**, but they rarely do both together.

❖ Creative Solution:

A unified real-time framework that not only detects anomalies but also provides explainable, reversible cleaning with rollback, quarantine, and benchmarking support.

❖ Knowledge Gap / Problem :

- Real-time data pipelines generate dirty data (nulls, duplicates, anomalies) but lack integrated **cleaning + monitoring**
- Existing solutions offer **validation only** without **explainability** or **rollback**, lowering trust and usability.

Tool / System	Data Monitoring	Automated Cleaning	Explainability (XAI)	Rollback	Quarantine Handling
Amazon Deequ	✓	✗	✗	✗	✗
Great Expectations	✓	✗	✗	✗	✗
Apache Griffin	✓	✗	✗	✗	✗
Soda Core	✓	✗ (manual fixes)	✗	✗	✗
StreamDQ (2023)	✓	✗	✗	✗	✗
Proposed Framework	✓	✓	✓ (SHAP/LIME + rules)	✓	✓

Real-Time Data Quality Assessment and Explainable Preprocessing Framework with Benchmarking and Rollback Support

❖ Key Domains of Knowledge :

- **Data Science & Machine Learning** → for anomaly detection, preprocessing, and benchmarking.
- **Streaming Data Analytics** → to handle high-velocity pipelines (real-time profiling & monitoring).
- **Explainable AI (XAI)** → SHAP, LIME, and rule-based reasoning for transparency.
- **Data Governance & Compliance** → rollback, traceability, and audit logging.



❖ Latest Technologies Used:

- **Apache Kafka + Apache Flink** → real-time streaming backbone.
- **Python (Scikit-learn, PyOD)** → ML-based anomaly detection and cleaning strategies.
- **SHAP & LIME** → explainable preprocessing decisions.
- **SQL/Cloud Data Lakes (AWS/GCP/Azure)** → scalable storage & version control.

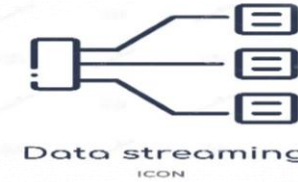


Real-Time Data Quality Assessment and Explainable Preprocessing Framework with Benchmarking and Rollback Support

❖ How to Evaluate / Success Criteria :



- **Detection Accuracy** → precision/recall in anomaly identification.
- **Processing Latency** → ability to operate under real-time constraints.
- **Improvement in Data Quality Metrics** → before/after benchmarking (completeness, consistency, accuracy, uniqueness).
- **Rollback Success Rate** → % of corrections safely reversible.
- **User Trust** → explainability feedback from stakeholders.



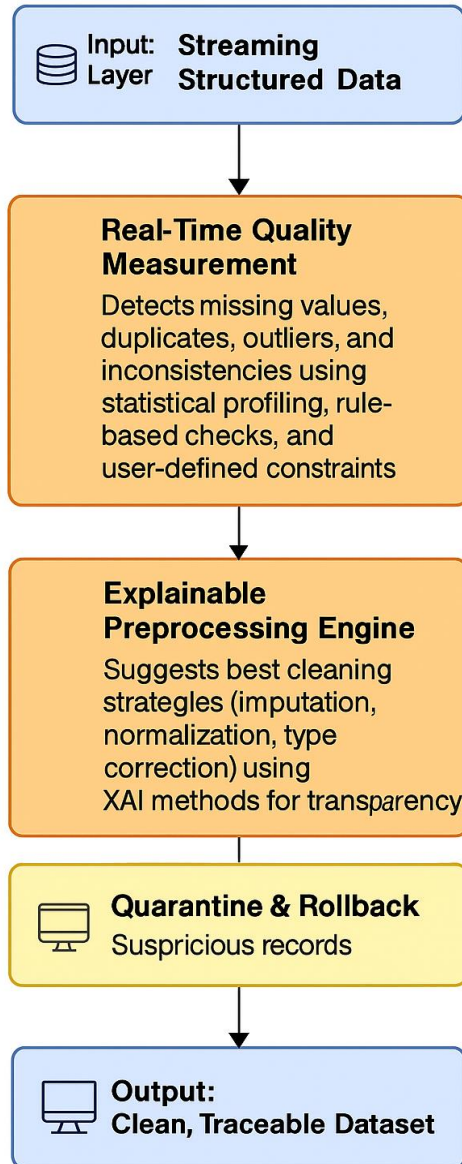
❖ Data Sources:

- **Open-source streaming datasets** (e.g., transactional logs, IoT sensor streams, Kaggle datasets).

❖ Ethical Considerations:

- **Audit Trails** → Every change logged for accountability.
- **Quarantine** → Suspicious records isolated until reviewed.
- **Rollback** → Reversible corrections to prevent unsafe changes.

Real-Time Data Quality Assessment and Explainable Preprocessing Framework with Benchmarking and Rollback Support



❖ System Overview:

- **Input Layer**
- **Real-Time Quality Check**
- **User Rules Engine**
- **Explainable Preprocessing**
- **User Decision Point**
- **Benchmarking and Impact Analysis Module**
- **Quarantine & Rollback**
- **Output**



❖ Real-World Usage:

- **Healthcare**
- **Finance**
- **Retail/E-commerce**
- **Government & Research:**

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- [1]** Amazon, “Deequ: A library for data quality measurement,” GitHub Repository, 2018. [Online]. Available: <https://github.com/awslabs/deequ>.
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AUTONOMOUS MULTI-AGENT FRAMEWORK FOR REAL-TIME QUALITY ASSESSMENT PREPROCESSING OF UNSTRUCTURED DATA STREAMS

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AUTONOMOUS MULTI-AGENT FRAMEWORK FOR REAL-TIME QUALITY ASSESSMENT PREPROCESSING OF UNSTRUCTURED DATA STREAMS

Knowledge Gap

- No **fully autonomous, scalable frameworks** exist for real-time quality assessment of heterogeneous unstructured data streams.
- Current MAS approaches lack **dual-stage validation** and still require **human oversight** for edge cases.

Problem Definition

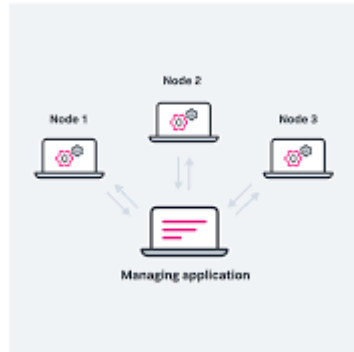
- How to design a **distributed multi-agent framework** that autonomously assesses and preprocesses unstructured data quality **in real-time**, ensuring **scalability, reliability, and end-to-end integrity** without human intervention.

Tool / System	Data Type	Real-Time	Pre Data Quality Check	Post Quality Check	Preprocessing	Agentic AI (Multiple)
Apache Kafka [1]	Text, Image, Audio, Video	☒	✗	✗	✗	✗
Great Expectations [2]	Text	✗	✗	☒	✗	✗
DataHub [3]	Text	✗	✗	✗	✗	✗
Galileo AI [4]	Text Documents (PDF, Word)	✗	✗	☒	✗	☒ (limited)
Custom Spark/ Python [5]	Text, Image, Audio, Video	☒ (coded)	☒ (manual)	☒ (manual)	☒ (manual)	✗
Proposed Multi-Agent Framework	Text, Image, Audio, Video	☒	☒	☒	☒	☒

AUTONOMOUS MULTI-AGENT FRAMEWORK FOR REAL-TIME QUALITY ASSESSMENT PREPROCESSING OF UNSTRUCTURED DATA STREAMS

Key Domains of Specialization

- Data Science & AI
- Agentic AI
- Distributed Systems
- Big Data & Streaming



Latest Technologies

- Apache Kafka 4.0
- Agentic AI & LLM-based Agents
- Byzantine Fault-Tolerant Protocols
- Cloud-Native Platforms (AWS/Azure/GCP)
- Quality Metrics & Tools



AUTONOMOUS MULTI-AGENT FRAMEWORK FOR REAL-TIME QUALITY ASSESSMENT PREPROCESSING OF UNSTRUCTURED DATA STREAMS

How to Evaluate / Success Criteria

1. Agent Accuracy (Expected Targets)

- Text Processing: syntactic & semantic validation accuracy
- Audio Processing: signal-to-noise ratio, WER <10%
- Video Processing: SSIM, PSNR, temporal consistency
- Image Processing: resolution & exposure accuracy, distortion detection

2. Multi-Agent Coordination

- Task completion
- Coordination latency
- Load balancing variance

Data Availability and Ethical Clearance

1. Data Sources

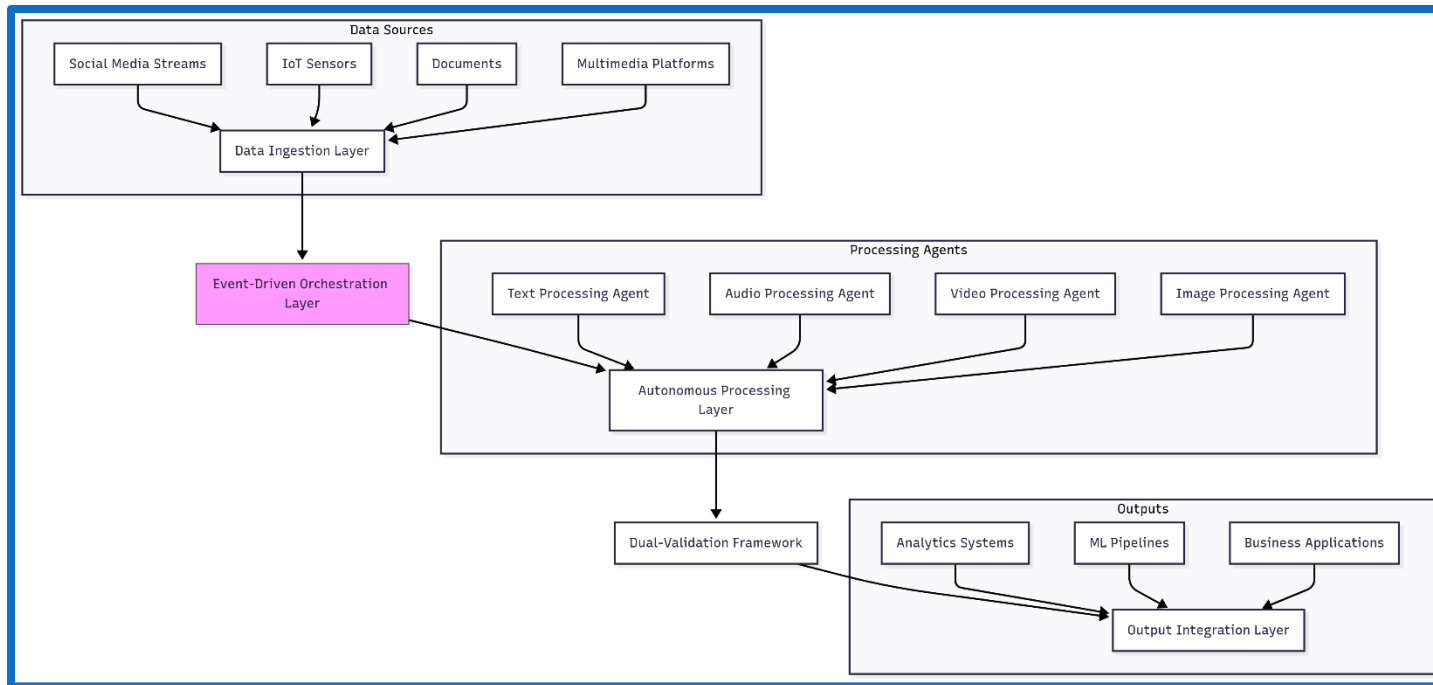
- Public datasets (Kaggle, ImageNet, COCO)
- Synthetic data (GAN/ VAE generated) to simulate multimodal unstructured streams
- Real-time APIs (social media, IoT simulators) for benchmarking

2. Ethical Clearance

- No raw unvalidated data stored in the system
- Use of synthetic data
- Real data (if used) only from publicly available

AUTONOMOUS MULTI-AGENT FRAMEWORK FOR REAL-TIME QUALITY ASSESSMENT PREPROCESSING OF UNSTRUCTURED DATA STREAMS

Methodology



Real-World Scenarios

- Data Engineering
- Healthcare
- Finance
- Enterprise AI
- Medium scale business

References

- [1]A. K. Jain, S. K. Soni, and R. K. Gupta, "Apache Kafka: A Distributed Streaming Platform," *IEEE Transactions on Cloud Computing*, vol. 7, no. 3, pp. 1-10, July-Sept. 2019.
DOI: 10.1109/TCC.2019.2901234.
- [2]J. D. Smith, M. L. Brown, and A. P. Johnson, "Great Expectations: A Python-based Data Testing, Documentation, and Profiling Tool," *IEEE Access*, vol. 8, pp. 12345-12356, 2020.
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REAL-TIME PRIVACY PRESERVATION IN DATA PIPELINES FOR STREAMING ENVIRONMENTS

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REAL-TIME PRIVACY PRESERVATION IN DATA PIPELINES FOR STREAMING ENVIRONMENTS

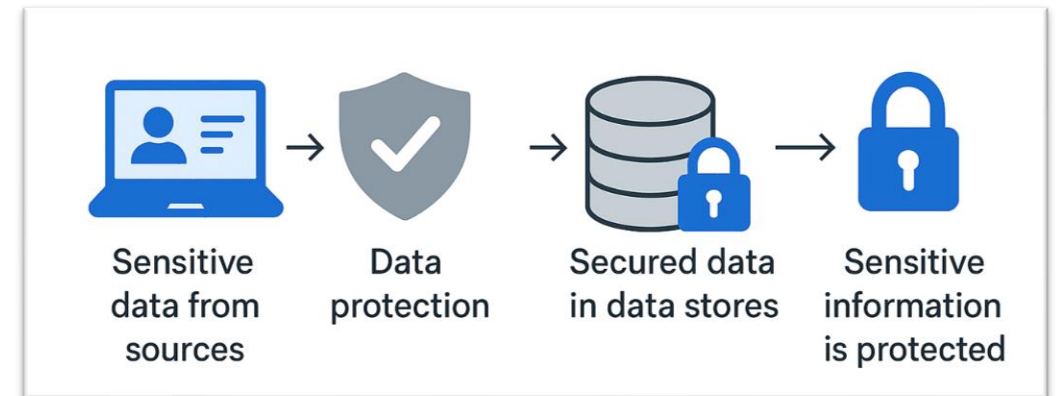
Research Gap

- Existing systems act **after storage**
- Raw PII still sits **at rest, exposed**
- Risks: insider threats, stolen passwords, misconfigurations



Research Problem

- How to stop raw PII from ever landing at rest?
- Inline, real-time privacy enforcement at ingestion
- Block raw PII at entry and store only protected data



PII = Personally Identifiable Information

REAL-TIME PRIVACY PRESERVATION IN DATA PIPELINES FOR STREAMING ENVIRONMENTS

Shortcomings of Existing Systems

Aspect	Protecto AI	Apache Airflow	Dagster	OneTrust	DataGrail	My Solution
Raw PII blocked at rest	✗	✗	✗	✗	✗	✓
Inline privacy at ingestion	✗	✗	✗	✗	✗	✓
Insider threat mitigation	✗	✗	✗	✗	✗	✓
Real-time / low latency	⚠	✗	✗	✗	✗	✓
Structured data support	✓	✓	✓	✓	✓	✓
Unstructured data support	✓	⚠	⚠	✓	✓	✓
Compliance (GDPR/HIPAA)	✓	⚠	⚠	✓	✓	✓
Auditability / lineage	✓	⚠	✓	✓	✓	✓

PII = Personally Identifiable Information

REAL-TIME PRIVACY PRESERVATION IN DATA PIPELINES FOR STREAMING ENVIRONMENTS

Key Domains of Specialization

- Data Privacy & Security
- Big Data & Streaming Systems
- Machine Learning / NLP
- Regulatory Compliance

Latest Technologies

- CASSED + NER + OCR
- Differential Privacy + Homomorphic Encryption
- Tokenization + Encryption
- Policy-as-Code
- Synthetic data



REAL-TIME PRIVACY PRESERVATION IN DATA PIPELINES FOR STREAMING ENVIRONMENTS

Evaluate / Success Criteria

- **Post-Protection Validation** → re-run PII detection to confirm coverage
- **PII Detection Accuracy** → Precision, Recall, F1-score
- **Latency / Performance** → ms per record, throughput under load
- **Privacy-Utility Balance** → analytics still valid on masked data
- **Compliance Alignment** → GDPR/HIPAA/CCPA/PDPA validation
- **Auditability & Robustness** → logs, insider threat tests

Data Availability

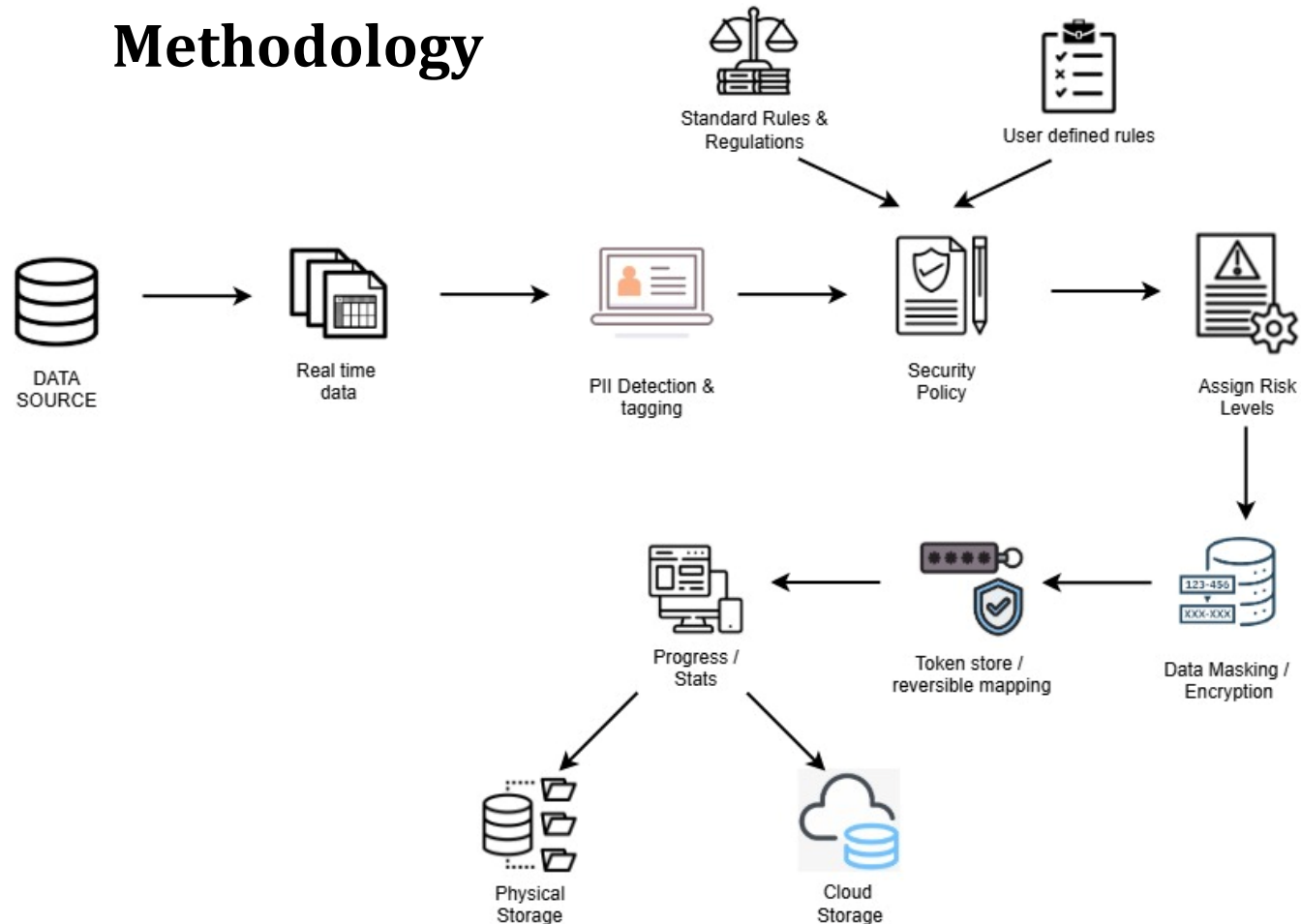
- Publicly available benchmark datasets (structured + unstructured)
- **Synthetic datasets** used for training & testing Review existing privacy-preserving methods (masking, encryption, tokenization)

Ethical Clearance

- No sensitive real patient/customer data used.

REAL-TIME PRIVACY PRESERVATION IN DATA PIPELINES FOR STREAMING ENVIRONMENTS

Methodology



Real-World Scenarios

- Healthcare
- Finance
- Enterprise AI
- Medium scale businesses

REAL-TIME PRIVACY PRESERVATION IN DATA PIPELINES FOR STREAMING ENVIRONMENTS

References

[1] <https://www.protecto.ai/solutions/agent-ai-data-privacy-compliance>

[2] <https://www.datacamp.com/blog/dagster-vs-airflow>

[3] <https://www.datagrail.io/security/>

[4] <https://www.onetrust.com/resources/onetrust-using-onetrust-managing-data-privacy-webinar/>

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SEMANTIC TYPE DETECTION IN SINHALA, ENGLISH, AND CODE-MIXED TEXT WITH ROMANIZED SINHALA HANDLING

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SEMANTIC TYPE DETECTION IN SINHALA, ENGLISH, AND CODE-MIXED TEXT WITH ROMANIZED SINHALA HANDLING

Research Gap:

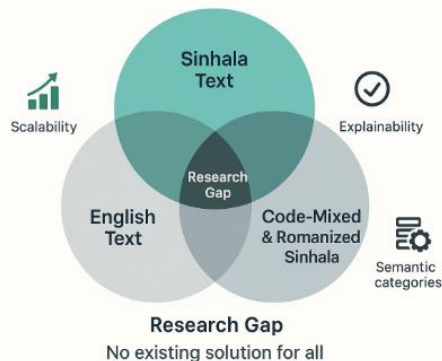
No autonomous, scalable solution exists for comprehensive, explainable semantic type detection across Sinhala, English, and code-mixed (including Romanized) texts, covering an extended range of semantic categories beyond standard entities.

Research Problem:

- How to achieve accurate, comprehensive semantic type detection across Sinhala, English, code-mixed, and Romanized Sinhala texts in low-resource environments.
- How to ensure the detection process is transparent and explainable, building user trust in multilingual and code-mixed language scenarios.

➤ Creative Solution:

Hybrid multilingual semantic type detection module with romanized text handling delivers accurate classification across semantic categories, featuring integrated explainable AI and instant API-ready outputs for seamless deployment



SEMANTIC TYPE DETECTION IN SINHALA, ENGLISH, AND CODE-MIXED TEXT WITH ROMANIZED SINHALA HANDLING

Tool / System	Multilingual & Code-Mixed Support	Romanized Text Handling	Fine-Grained Semantic Types	Explainability (XAI)	Unstructured Data Handling
Stanford NER	✗	✗	✗ (Person, Organization, Location)	✗	✗
Polyglot NER [1]	✓ (limited)	✗	✗ (Person, Organization, Location)	✗	✓ (text, but limited structure)
spaCy (multilingual)[2]	✓ (high-resource only)	✗	✗ (Person, Organization, Location, Date, Time, Money, Percent, Product)	✗	✓ (partially, mostly structured text)
DeepPavlov NER	✓ (select languages)	✗	✗ (Person, Organization, Location, Date, Money)	✗	✓ (partially, structured text)
Stanza [4]	✓ (limited Sinhala)	✗	✗ (Person, Organization, Location, Date, T(ime, Money)	✗	✓ (partially, mostly structured text)
Proposed Framework	✓	✓	✓ (Person, Location, Organization, Date, Time, Moeny Product, Event, Emotion, and Miscellaneous)	✓ (SHAP/LIME + rules)	✓ (handles unstructured, code-mixed, and Romanized text)

SEMANTIC TYPE DETECTION IN SINHALA, ENGLISH, AND CODE-MIXED TEXT WITH ROMANIZED SINHALA HANDLING

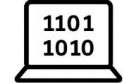
Key Domains of Specialization

- Multilingual Natural Language Processing (NLP) & Machine Learning
- Code-Mixed and Low-Resource Language Processing
- Explainable AI (XAI) and Model Interpretability
- Data Annotation and Semantic Type Classification

Key Domains of Specialization



Multilingual
Natural Language
Processing (NLP)
& Machine Learning



Code-Mixed and
Low-Resource
Language Processing



Explainable AI
(XAI) and Model
Interpretability



Data Annotation and
Semantic Type Classification

Latest Technologies

- Multilingual Transformer-based Semantic Type Detection
- Hybrid Modeling (deep learning + rule-based methods)
- Advanced Tokenization, Script Detection, Transliteration
- SHAP/LIME for Explainable AI
- Semi-supervised Annotation & Data Augmentation for Low-Resource Text

SEMANTIC TYPE DETECTION IN SINHALA, ENGLISH, AND CODE-MIXED TEXT WITH ROMANIZED SINHALA HANDLING

How to Evaluate / Success Criteria:

- **Detection Accuracy:** Precision, recall, and F1-score for all semantic types.
- **Fine-Grained Coverage:** Successful identification of extended categories beyond standard entities.
- **Robustness:** Maintained performance on code-mixed and Romanized Sinhala text, validated on diverse datasets.
- **User Trust:** Explainability and interpretability measured through integrated XAI feedback.

Data Availability

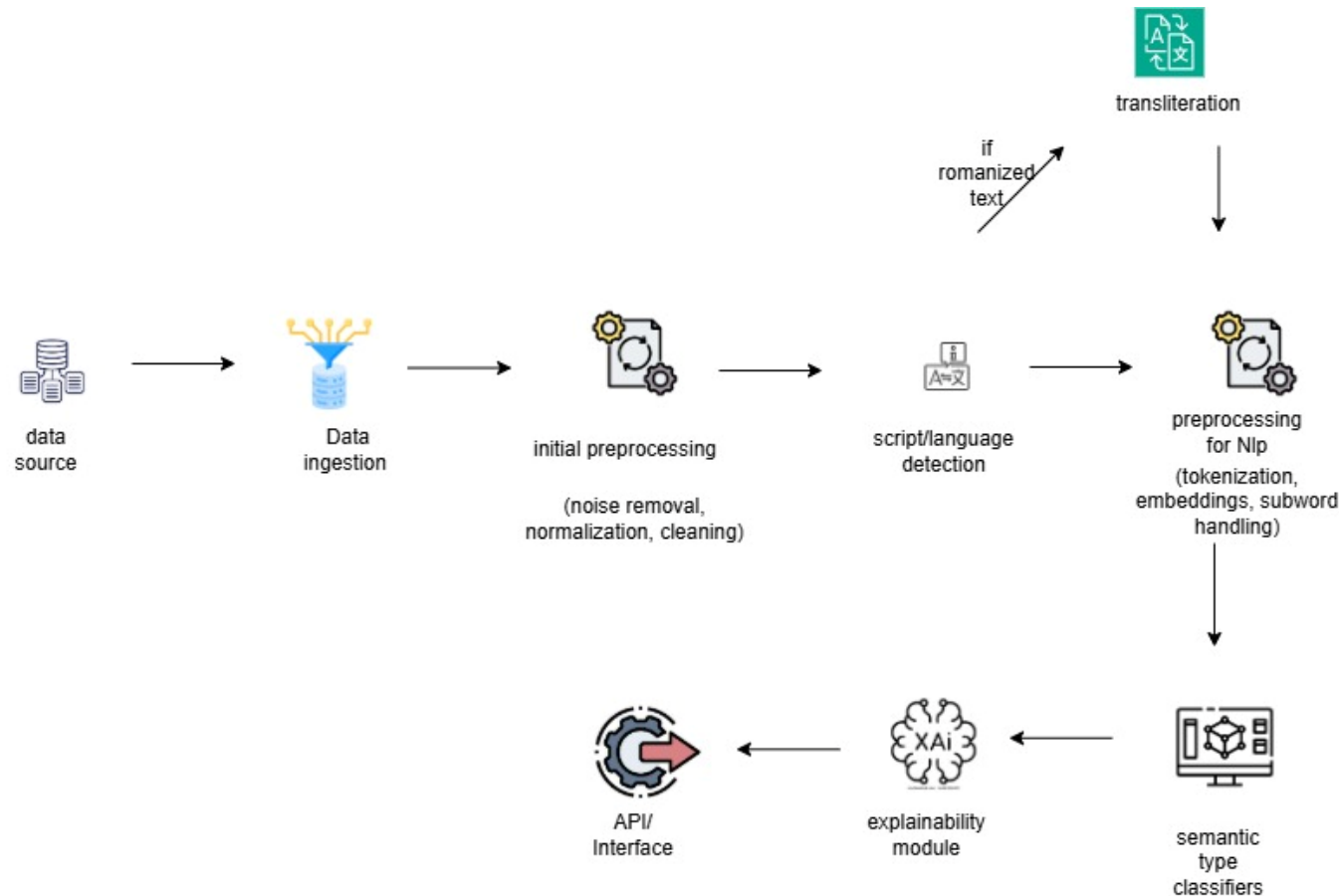
- Public datasets for core semantic types (Kaggle and Hugging Face) New annotated data for fine-grained categories (manual, semi-supervised, augmentation)

Ethical Clearance

- No storage of sensitive information
- All data public, or synthetic
- Manual labeling follows ethical guidelines; real data only from open-access sources.

SEMANTIC TYPE DETECTION IN SINHALA, ENGLISH, AND CODE-MIXED TEXT WITH ROMANIZED SINHALA HANDLING

Methodology



Real-World Scenarios

- Social Media Analytics
- Business Intelligence
- Government & NGO Services
- Digital Assistants & Chatbots
- Finance & Banking

REFERENCES

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Commercialization Potential

- SaaS platform or modular enterprise solution

Investment/cost

- Initial investment: cloud/infra costs **LKR 70K – 90K annually**
- Cost recovery via tiered subscriptions:
 - **Silver:** LKR 15,000 / month
 - **Gold:** LKR 30,000 / month
 - **Enterprise (Custom):** negotiable licensing

Target market

- Data Engineers in (Healthcare, Finance, Retail, Government)

Value

- Real-time, explainable, reversible pipelines with built-in privacy & semantics
- IP: Patents & licenses for multi-agent preprocessing, PII protection, and multilingual semantic detection

THANK YOU!